

MATH 345 Skeleton Notes

Unit 2: Anatomy of a linear map

Section 2.4: Independence, bases, and rank

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Linear independence

Activity: Comparing two spanning sets (U2-LO4)

In A set described by two parameters (U2-LO4) we showed that the set

$$W = \left\{ \begin{bmatrix} r \\ s \\ r + 2s \end{bmatrix} : r, s \in \mathbb{R} \right\}$$

is a subspace. In Writing a parametric set as a span (U2-LO4), we showed that the set

$$S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \right\}$$

is a spanning set for W . The set

$$T = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix} \right\}$$

also spans W . But why is S ‘better’?

It uses fewer vectors to span the same space.
Which vectors in T are redundant?

$$\begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}.$$

So T spans W , but it contains repeated information.

A subspace U has many different spanning sets. We use the concept of *linear independence* to determine when sets of spanning vectors are the ‘most efficient’. In data terms, independence asks whether one feature direction is already built from the others.

Definition: Linear Independence

A set of vectors $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ in $\mathbb{R}^n$ is said to be

linearly independent

, or simply

independent

if the only solution to the equation $t_1\mathbf{x}_1 + \dots + t_k\mathbf{x}_k = \mathbf{0}$ is trivial, i.e., $t_1 = \dots = t_k = 0$.

A set of vectors $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ in $\mathbb{R}^n$ is

linearly dependent

if it

is not linearly independent, i.e., if there exists scalars $t_1, \dots, t_k$, not all zero, such that $t_1\mathbf{x}_1 + \dots + t_k\mathbf{x}_k = \mathbf{0}$.

Example: Testing independence in three dimensions (U2-LO4)

Determine whether the vectors

$$\mathbf{v}_1 = \begin{bmatrix} 3 \\ 2 \\ 2 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix} \quad \mathbf{v}_3 = \begin{bmatrix} -1 \\ 4 \\ -2 \end{bmatrix}$$

are linearly independent.

We need to determine the solutions to the equation $t_1\mathbf{v}_1 + t_2\mathbf{v}_2 + t_3\mathbf{v}_3 = \mathbf{0}$. Expanding out this equation gives the system

$$\begin{aligned} 3t_1 + t_2 - t_3 &= 0 \\ 2t_1 + 3t_2 + 4t_3 &= 0 \\ 2t_1 + 0t_2 - 2t_3 &= 0. \end{aligned}$$

The Elimination Scratchpad

We solve this homogeneous system of linear equations using the Gaussian elimination algorithm. First we create the augmented matrix

$$\left[\begin{array}{ccc|c} 3 & 1 & -1 & 0 \\ 2 & 3 & 4 & 0 \\ 2 & 0 & -2 & 0 \end{array} \right]$$

We reduce the matrix using row operations to row echelon form:

$$\begin{aligned}
 & \left[\begin{array}{ccc|c} 3 & 1 & -1 & 0 \\ 2 & 3 & 4 & 0 \\ 2 & 0 & -2 & 0 \end{array} \right] \\
 & \rightarrow \left[\begin{array}{ccc|c} 3 & 1 & -1 & 0 \\ 0 & \frac{7}{3} & \frac{14}{3} & 0 \\ 0 & -\frac{2}{3} & -\frac{4}{3} & 0 \end{array} \right] \\
 & \rightarrow \left[\begin{array}{ccc|c} 3 & 1 & -1 & 0 \\ 0 & \frac{7}{3} & \frac{14}{3} & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]
 \end{aligned}$$

There is one free variable corresponding to this matrix, which implies that there is one basic solution to the equation, and thus the system has a nontrivial solution. Thus the vectors are *not* linearly independent.

The last example demonstrates an algorithm to verify whether a given set of vectors is linearly independent.

Algorithm: A Test of Independence

To verify if a set $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ are linearly independent:

1. Consider the system of linear equations $t_1\mathbf{x}_1 + \dots + t_k\mathbf{x}_k = \mathbf{0}$.

2. Solve the resulting linear homogeneous system.
3. If there is a nontrivial solution to the equation, then the vectors are linearly dependent. Otherwise, the vectors are linearly independent.

In matrix form: put the vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$ as the columns of a matrix M . Then $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ is independent exactly when $M\mathbf{c} = \mathbf{0}$ has only the trivial solution, equivalently when every column of M is a pivot column.

Theorem

If $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ is a linearly independent set of vectors in $\mathbb{R}^n$, then every vector in their span, $\text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$, has a unique representation as a linear combination of the $\mathbf{x}_i$.

Why is this true?

Let $\mathbf{x}$ be an element of $\text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$. If the equation

$$\mathbf{x} = a_1\mathbf{x}_1 + \dots + a_k\mathbf{x}_k$$

had another solution, i.e.,

$$\mathbf{x} = b_1\mathbf{x}_1 + \dots + b_k\mathbf{x}_k,$$

then subtracting one equation from the other gives that

$$(a_1 - b_1)\mathbf{x}_1 + \dots + (a_k - b_k)\mathbf{x}_k = \mathbf{0}.$$

Thus the coefficients $a_1 - b_1, \dots, a_k - b_k$ give a solution to the equation

$$t_1\mathbf{x}_1 + \dots + t_k\mathbf{x}_k = \mathbf{0}.$$

But the vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$ are linearly independent, and so this equation has only a trivial solution. Thus $a_1 - b_1, \dots, a_k - b_k = 0$, i.e., $a_1 = b_1, \dots, a_k = b_k$, proving that the representation of $\mathbf{x}$ as a linear combination is unique.

Activity: Testing independence in four dimensions (U2-LO4)

Determine whether the vectors

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ -2 \\ 0 \\ 0 \end{bmatrix} \quad \mathbf{v}_3 = \begin{bmatrix} 0 \\ -3 \\ -1 \\ -1 \end{bmatrix}$$

are linearly independent.

We must check if the equation

$$t_1 \mathbf{v}_1 + t_2 \mathbf{v}_2 + t_3 \mathbf{v}_3 = \mathbf{0}$$

has nontrivial solutions. Expanding out this equation gives the system

$$t_1 + t_2 = 0$$

$$2t_1 - 2t_2 - 3t_3 = 0$$

$$-t_3 = 0$$

$$t_1 - t_3 = 0$$

The Elimination Scratchpad

We could use row-reduction here, though in this case a more ad-hoc approach is more efficient. The third equation gives $t_3 = 0$, and once this is

deduced the fourth equation gives $t_1 = 0$. Then the first equation gives $t_2 = 0$. Thus the only solution to the equation is the trivial solution, and so the vectors are linearly independent.

Fact

The nonzero rows of a matrix in row echelon form are linearly independent.

Basis and dimension

In geometry, we often describe lines as being ‘1-dimensional’, and planes as being ‘2-dimensional’. We can describe the notion of dimension more precisely using linear algebra.

Theorem: A Fundamental Theorem

Let U be a subspace of $\mathbb{R}^n$. If U is spanned by m vectors, and if U contains k linearly independent vectors, then $k \leq m$.

Activity: Four vectors in three dimensions (U2-L04)

Is it possible to have 4 linearly independent vectors in $\mathbb{R}^3$?

It is not possible. $\mathbb{R}^3$ is spanned by 3 vectors (for example, the standard basis vectors), so by A Fundamental Theorem, any linearly independent subset of $\mathbb{R}^3$ must contain at most 3 vectors.

Definition: Bases of Vector Spaces

If U is a subspace of $\mathbb{R}^n$, a set of vectors $\{\mathbf{x}_1, \dots, \mathbf{x}_m\}$ in U is called

a **basis** of U if it satisfies the following two conditions:

- $\{\mathbf{x}_1, \dots, \mathbf{x}_m\}$ is *linearly independent*.
- $U = \text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_m\}$.

Theorem: The Invariance Theorem

If $\{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ and $\{\mathbf{w}_1, \dots, \mathbf{w}_k\}$ are both *bases* for a subspace U of $\mathbb{R}^n$, then $m = k$.

The Invariance Theorem says that, even though a subspace has many different bases, all bases must have the *same number* of vectors.

Definition: Dimensions of Vector Spaces

The **dimension** of a non-zero subspace U of $\mathbb{R}^n$ is the *number* of vectors in a *basis* for U . We often write $\dim(U)$ for the dimension of U , and if $\dim(U) = d$, we say U is d -dimensional. By convention, the dimension of the trivial subspace $\{\mathbf{0}\}$ is *zero*.

- **Term:** spanning set

Meaning: enough directions to generate the whole subspace

- **Term:** independent set
Meaning: no direction is redundant
- **Term:** basis
Meaning: just enough directions: spanning and independent
- **Term:** dimension
Meaning: number of directions in any basis

Activity: Dimensions of coordinate spaces (U2-LO4)

What is the dimension of $\mathbb{R}^3$?

The standard basis for $\mathbb{R}^3$ is a basis, i.e., the set of three vectors $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ defined in Standard basis is linearly independent and spans $\mathbb{R}^3$. Thus $\mathbb{R}^3$ has dimension 3.

Activity: Dimensions of coordinate spaces (U2-LO4)

What is the dimension of $\mathbb{R}^4$?

The standard basis $\mathbf{e}_1, \dots, \mathbf{e}_4$ for $\mathbb{R}^4$ is a basis, and is a set of four vectors, so $\dim(\mathbb{R}^4) = 4$.

Activity: Dimensions of coordinate spaces (U2-LO4)

What is the dimension of $\mathbb{R}^n$?

The standard basis is a basis for $\mathbb{R}^n$, so $\dim(\mathbb{R}^n) = n$.

Activity: The dimension of a parametric subspace (U2-LO4)

Find the dimension of the space

$$W = \left\{ \begin{bmatrix} r \\ s \\ r + 2s \end{bmatrix} : r, s \in \mathbb{R} \right\}.$$

The set

$$S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \right\}$$

is a basis for W since S is linearly independent (neither vector in S is the scalar multiple of the other). In the solution to Writing a parametric set as a span (U2-LO4), we showed that these

vectors span W , so S is a basis for W . Since S contains two vectors, $\dim(W) = 2$.

Fact

Let U be a non-zero subspace of $\mathbb{R}^n$. Then

- U has a basis and $\dim(U) \leq n$.
- Any independent set in U can be enlarged (by adding some subset of vectors in any fixed basis) to obtain a basis of U .
- Any spanning set for U contains a subset of vectors which form a basis for U .

Theorem

Let U be a subspace of $\mathbb{R}^n$ where $\dim(U) = m$ and let $B = \{\mathbf{x}_1, \dots, \mathbf{x}_m\}$ be a set of m vectors in U . Then B is linearly independent if and only if B spans U .

Why is this true?

Suppose B is linearly independent and $|B| = m = \dim(U)$. By A Fundamental Theorem, one cannot add any new vectors to the set B and remain linearly independent. But the referenced item says that if B is not already a spanning set for U , one can add new vectors to B from U , and remain linearly independent. Thus B must be a basis.

Conversely, if B spans U , then by the referenced item we can choose a subset of B which is a basis for U . But any proper subset of B contains fewer than m vectors, and thus by The Invariance Theorem, cannot be a basis for U . Thus B is the only subset of itself that can be a basis, and because B must contain a basis, the set B must itself be a basis.

Theorem

Let U and W be subspaces of $\mathbb{R}^n$ with $U \subseteq W$. Then:

- $\dim(U) \leq \dim(W)$.
- If $\dim(U) = \dim(W)$, then $U = W$.

Activity: Dependence and the geometry of two vectors (U2-LO4)

If $\mathbf{v}$ and $\mathbf{w}$ are nonzero vectors in $\mathbb{R}^3$, show that $\{\mathbf{v}, \mathbf{w}\}$ is dependent if and only if $\mathbf{v}$ and $\mathbf{w}$ are parallel, i.e., if one vector is a scalar multiple of the other.

If $\mathbf{v}$ and $\mathbf{w}$ are parallel, then one is a scalar multiple of the other, say $\mathbf{v} = a\mathbf{w}$ for some scalar a . Then $1\mathbf{v} - a\mathbf{w} = \mathbf{0}$, which is a nontrivial linear combination of $\mathbf{v}$ and $\mathbf{w}$, so $\{\mathbf{v}, \mathbf{w}\}$ is dependent. Conversely, if $\{\mathbf{v}, \mathbf{w}\}$ is dependent, then $s\mathbf{v} + t\mathbf{w} = \mathbf{0}$ for some scalars s and t , where $s \neq 0$ or $t \neq 0$. If, say, $s \neq 0$, then $\mathbf{v} = -t/s\mathbf{w}$, so $\mathbf{v}$ and $\mathbf{w}$ are parallel. A similar argument works if $t \neq 0$.

Activity: Adding a vector to an independent pair (U2-LO4)

Let $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are non-zero vectors in $\mathbb{R}^n$, where $\{\mathbf{v}, \mathbf{w}\}$ is independent. Show that $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is independent if and only if $\mathbf{u}$ is not in the subspace $M = \text{span}\{\mathbf{v}, \mathbf{w}\}$. See the figure and the figure.

Activity: Adding a vector to an independent pair (U2-LO4) (continued)

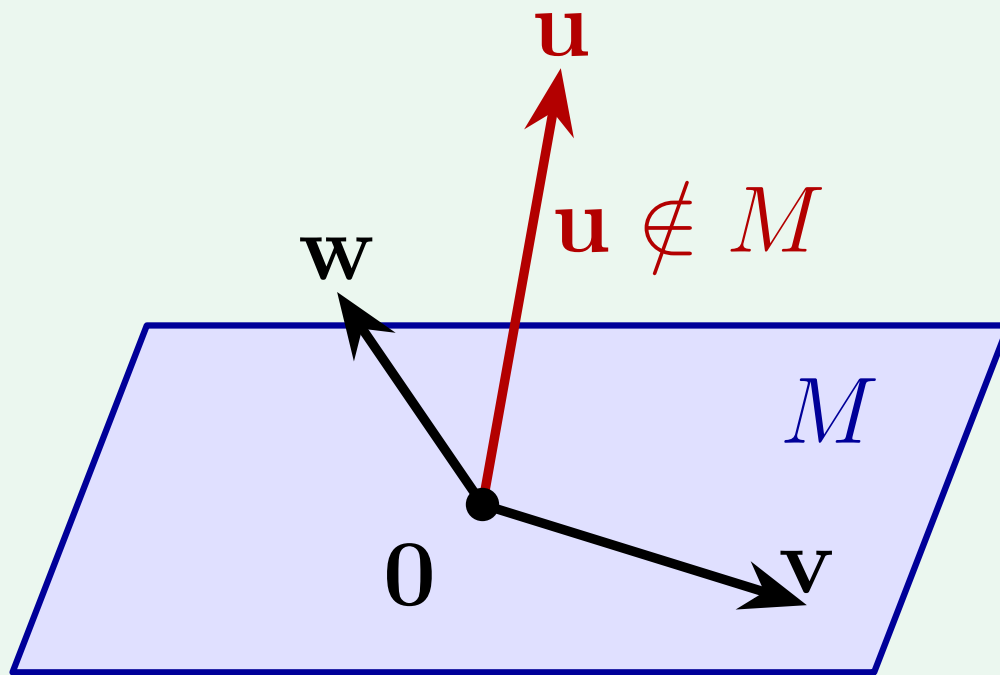


Figure. $\{u, v, w\}$ independent.

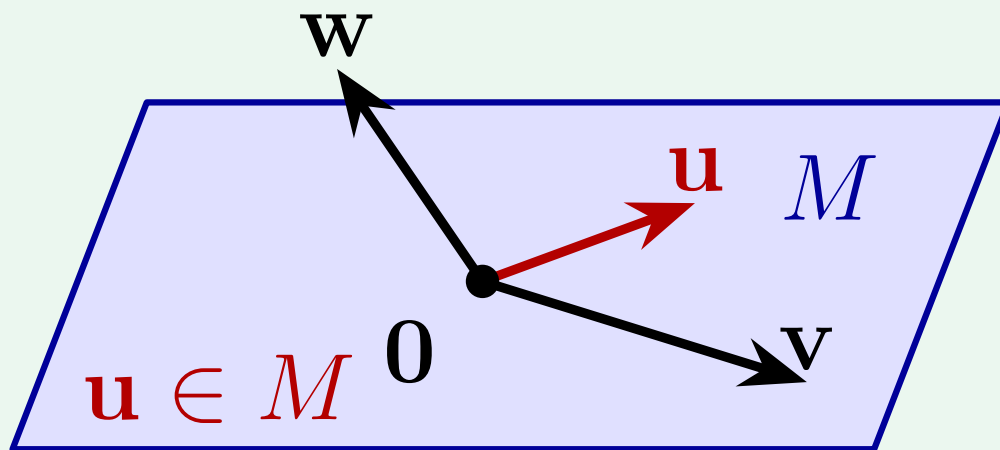


Figure. $\{u, v, w\}$ dependent.

If $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is independent, suppose that $\mathbf{u}$ is in $M = \text{span}\{\mathbf{v}, \mathbf{w}\}$, say, that $\mathbf{u} = a\mathbf{v} + b\mathbf{w}$ for some scalars a and b . Then $1\mathbf{u} - a\mathbf{v} - b\mathbf{w} = \mathbf{0}$. Thus a nontrivial linear combination of $\mathbf{u}, \mathbf{v}$, and $\mathbf{w}$ is equal to zero, contradicting the independence of $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$.

On the other hand, suppose $\mathbf{u}$ is not in M . We need to show that $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is independent. If $r\mathbf{u} + s\mathbf{v} + t\mathbf{w} = \mathbf{0}$, where r, s , and t are not all zero, then $r \neq 0$, for otherwise $s\mathbf{v} + t\mathbf{w} = \mathbf{0}$, which contradicts the independence of $\{\mathbf{v}, \mathbf{w}\}$. But then $\mathbf{u} = -s/r\mathbf{v} - t/r\mathbf{w}$, which contradicts that $\mathbf{u}$ is not in M .

Note

More generally, if $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ are a linearly independent set of vectors in $\mathbb{R}^n$, and $\mathbf{x}_{k+1}$ is another vector in $\mathbb{R}^n$, then $\{\mathbf{x}_1, \dots, \mathbf{x}_{k+1}\}$ is a linearly independent set if and only if $\mathbf{x}_{k+1}$ is not contained in the span of the set $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$.

Activity: Extending a pair of vectors to a basis (U2-L04)

Find a basis for $\mathbb{R}^4$ that contains the two vectors

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 2 \\ 0 \end{bmatrix} \quad \text{and} \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 0 \\ 3 \\ 3 \end{bmatrix}.$$

Since $\mathbb{R}^4$ is four dimensional, we must find two vectors $\mathbf{w}_1$ and $\mathbf{w}_2$ such that the vectors $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{w}_1, \mathbf{w}_2\}$ is linearly independent. The referenced item states that we can choose these vectors from any basis, so we might as well choose them from the standard basis $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_4\}$.

One strategy is to choose one vector $\mathbf{e}_i$ not contained in the span of the set $\{\mathbf{v}_1, \mathbf{v}_2\}$, and then choose a vector $\mathbf{e}_j$ not contained in the span of the set $\{\mathbf{e}_i, \mathbf{v}_1, \mathbf{v}_2\}$. By the referenced item, the set $\{\mathbf{e}_j, \mathbf{e}_i, \mathbf{v}_1, \mathbf{v}_2\}$ is then linearly independent. The vector $\mathbf{e}_2$ is not contained in the span of the set $\{\mathbf{v}_1, \mathbf{v}_2\}$, since the second entry in any linear combination of the vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ is equal to

zero, and thus the vector cannot be equal to $\mathbf{e}_2$. We also claim that the vector $\mathbf{e}_1$ is not contained in the span of the set $\{\mathbf{e}_2, \mathbf{v}_1, \mathbf{v}_2\}$. If

$$\mathbf{e}_1 = a\mathbf{e}_2 + b\mathbf{v}_1 + c\mathbf{v}_2,$$

then checking each entry, we get the system of linear equations

$$1 = b$$

$$0 = a$$

$$0 = 2b + 3c$$

$$0 = 3c$$

The Elimination Scratchpad

The first, third, and fourth equations immediately contradict one another, so such a system cannot be solved.

Thus $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{v}_1, \mathbf{v}_2\}$ is a basis for $\mathbb{R}^4$.

Bases and coordinate systems

One way to think of a basis is as providing a coordinate system for a subspace. The uniquely determined coefficients in the linear combination of a vector in the subspace are the coordinates.

Definition: Ordered Bases of a Subspace

Let $S = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ be a basis of the subspace U . We call the tuple $B = (\mathbf{b}_1, \dots, \mathbf{b}_n)$ an **ordered basis** of U .

As an example, $(\mathbf{e}_1, \mathbf{e}_2)$ and $(\mathbf{e}_2, \mathbf{e}_1)$ are two different ordered bases of $\mathbb{R}^2$. The two sets $\{\mathbf{e}_1, \mathbf{e}_2\}$ and $\{\mathbf{e}_2, \mathbf{e}_1\}$ are equal, and so describe the same basis of $\mathbb{R}^2$.

Given an ordered basis B for an m -dimensional subspace U of $\mathbb{R}^n$, and any vector $\mathbf{v} \in U$, we can describe its ‘coordinates’ with respect to the basis B , identifying $\mathbf{v}$ with a vector in $\mathbb{R}^m$.

Definition: Coordinate Vectors With Respect to a Basis

Let U be an m -dimensional subspace of $\mathbb{R}^n$, and let $B = (\mathbf{b}_1, \dots, \mathbf{b}_m)$ be an ordered basis for U . Given any vector $\mathbf{v} \in U$, there exists unique scalars $t_1, \dots, t_m$ such that

$$\mathbf{v} = t_1 \mathbf{b}_1 + \dots + t_m \mathbf{b}_m$$

and we define the **coordinate vector** of a vector $\mathbf{v} \in U$ with respect to B to be the m -vector

$$C_B(\mathbf{v}) = \begin{bmatrix} t_1 \\ \vdots \\ t_m \end{bmatrix}.$$

Note

If $B = (\mathbf{b}_1, \dots, \mathbf{b}_m)$ is an ordered basis for a subspace U , then $C_B(\mathbf{b}_i) = \mathbf{e}_i$, where $\mathbf{e}_i$ is the i th standard basis vector in $\mathbb{R}^m$.

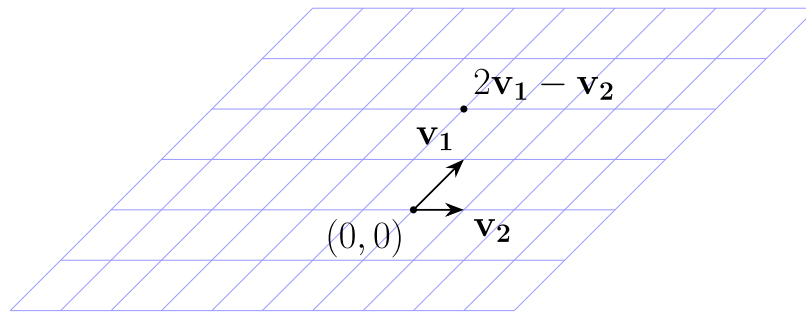


Figure. A slanted parallelogram grid determined by basis vectors $\mathbf{v}_1$ and $\mathbf{v}_2$.

Activity: Coordinates relative to an ordered basis (U2-LO4)

Let $B = (\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ be an ordered basis for $\mathbb{R}^3$, where

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad \mathbf{v}_3 = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}.$$

What is $C_B(\mathbf{v})$ if

$$\mathbf{v} = \begin{bmatrix} 1 \\ 1 \\ -5 \end{bmatrix}.$$

We need to find scalars c_1, c_2 , and c_3 such that

$$\mathbf{v} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + c_3 \mathbf{v}_3.$$

Comparing entries of the vectors gives a system of three equations, i.e.,

$$\begin{aligned}c_1 + c_2 + 2c_3 &= 1 \\c_2 &= 1 \\c_3 &= -5\end{aligned}$$

The Elimination Scratchpad

Using back substitution, we find immediately that $c_3 = -5$ from the last equation, then $c_2 = 1$ from the second. Substituting these values into the first equation gives

$$c_1 = 1 - c_2 - 2c_3 = 1 - 1 - 2(-5) = 10.$$

These values give the unique solution to this system. Thus

$$C_B(\mathbf{v}) = \begin{bmatrix} 10 \\ 1 \\ -5 \end{bmatrix}.$$

Column and row space

In this section, we will often realize vectors in $\mathbb{R}^n$ as rows rather than columns. The notion of span, linear independence, and basis are defined

analogously as how they are defined for column vectors.

Definition: Column and Row Space

Let

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

be an $m \times n$ matrix.

- The columns of A , considered as vectors in $\mathbb{R}^m$, span a *subspace* of $\mathbb{R}^m$ called the **column space** of A , denoted $\text{col}(A)$.
- The rows of A , considered as vectors in $\mathbb{R}^n$, span a *subspace* of $\mathbb{R}^n$ called the **row space** of A , denoted $\text{row}(A)$.

Note

The column space is the image space from Subspaces written with a column-based name:

$$\text{col}(A) = \text{im}(A).$$

The notation $\text{im}(A)$ emphasizes outputs of the map $\mathbf{x} \mapsto A\mathbf{x}$. The notation $\text{col}(A)$ emphasizes the columns that span those outputs.

Lemma

Let A and B be two row equivalent $m \times n$ matrices. Then

$$\text{row}(A) = \text{row}(B).$$

Why is this true?

Each time we apply a row operation to a matrix, we replace one or more

of its rows with a linear combination of other rows. Thus the row space of a matrix A' obtained by applying a row operation to a matrix A must be contained in the row space of A . But applying the argument in reverse (since every row operation is reversible), the row space of A must be contained in the row space of A' . Thus the two row spaces are equal. But if the row space is not changed after applying a single row operation, it is not changed after applying an arbitrary number of row operations, and so the result follows.

Lemma

If R is a matrix in row echelon form (recall Row echelon form and reduced row echelon form) then the non-zero rows of R are a basis of $\text{row}(R)$.

Why is this true?

The nonzero rows of R are linearly independent by the referenced item, and span $\text{row}(R)$ by definition. They thus by definition form a basis for $\text{row}(R)$.

Note

If A is row equivalent to a row echelon matrix R , then $\text{row}(A) = \text{row}(R)$. By the referenced item, the nonzero rows of R form a basis for this space. Thus $\dim(\text{row}(A))$ equals the number of nonzero rows, or equivalently the number of pivots, in R . In particular, this pivot count is independent of the choice of row echelon form.

Activity: Finding a basis for a span (U2-LO4)

Find a basis for $V = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$, where

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix} \quad \mathbf{v}_3 = \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix} \quad \mathbf{v}_4 = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}.$$

The subspace V is the row space of the matrix

$$\begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & -3 \\ 0 & 1 & -2 \\ 1 & 1 & -1 \end{bmatrix}.$$

If we reduce this matrix to row echelon form, we thus find a basis for V :

$$\begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & -3 \\ 0 & 1 & -2 \\ 1 & 1 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & \frac{1}{2} & 0 \\ 0 & \frac{3}{2} & -3 \\ 0 & 1 & -2 \\ 0 & \frac{1}{2} & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & \frac{1}{2} & 0 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Thus

$$\left\{ \begin{bmatrix} 1 \\ \frac{1}{2} \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix} \right\}$$

is a basis for V .

Theorem: Rank Theorem

Let A denote any $m \times n$ matrix. Then

$$\dim(\text{col}(A)) = \dim(\text{row}(A)).$$

Write r for this common dimension. If A is row-equivalent to a matrix R in row echelon form, then

- The r leading rows of R are a basis for $\text{row}(A)$.
- If the pivots of R lie in columns

$$j_1, j_2, \dots, j_r,$$

then the columns $j_1, j_2, \dots, j_r$ of A are a basis for $\text{col}(A)$.

Warning: Pivot columns come from the original matrix

Row reduction identifies pivot positions, but row operations do not preserve the original columns of A . When finding a basis for $\text{col}(A)$, use the pivot columns of the original matrix A , not the columns of the row-reduced matrix.

Definition: Rank of a matrix

The **rank** of a matrix A , denoted $\text{rank}(A)$, is the common dimension of its column and row spaces:

$$\text{rank}(A) = \dim(\text{col}(A)) = \dim(\text{row}(A)).$$

- **Goal:** basis for $\text{row}(A)$
Row reduction tells you: nonzero rows of REF/RREF
Use these vectors: rows of the reduced matrix
- **Goal:** basis for $\text{col}(A)$
Row reduction tells you: pivot column positions
Use these vectors: corresponding columns of original A

- **Goal:** basis for $\text{null}(A)$
Row reduction tells you: free variables
Use these vectors: basic solution vectors in $\mathbb{R}^n$
- **Goal:** rank
Row reduction tells you: number of pivots
Use these vectors: a number
- **Goal:** nullity
Row reduction tells you: number of free variables
Use these vectors: a number

Fact

It follows from Rank Theorem that for any matrix A , $\text{rank}(A) = \text{rank}(A^T)$. Indeed, the columns of A^T are the rows of A written as column vectors, so their spans have the same dimension. Hence

$$\text{rank}(A^T) = \dim(\text{col}(A^T)) = \dim(\text{row}(A)) = \text{rank}(A).$$

Theorem: Computing rank by row reduction

The rank of a matrix M is the number of pivots in any row echelon form row equivalent to M .

Why is this true?

By the referenced item, the pivot count equals $\dim(\text{row}(M))$, which is $\text{rank}(M)$ by Rank of a matrix.

This gives an algorithm for computing rank: use The Gaussian Elimination Algorithm to row-reduce the matrix to REF, then count the pivots.

Activity: Finding the rank of a matrix (U2-LO3)

Find the rank of the matrix

$$C = \begin{bmatrix} 1 & 1 & 2 & 7 \\ 0 & 1 & 0 & 3 \\ 2 & 1 & 4 & 11 \end{bmatrix}$$

We convert the matrix to row echelon form using The Gaussian Elimination Algorithm. Subtract 2 times Row 1 from Row 3:

$$\begin{bmatrix} 1 & 1 & 2 & 7 \\ 0 & 1 & 0 & 3 \\ 0 & -1 & 0 & -3 \end{bmatrix}$$

Add Row 2 to Row 3:

$$\begin{bmatrix} 1 & 1 & 2 & 7 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The matrix is now in REF with 2 leading ones, and so the rank of C is 2.

Activity: Finding the rank of a matrix (continued) (U2-LO3)

Find the rank of the matrix

$$D = \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 3 \\ 2 & 1 & 2 & -1 \end{bmatrix}$$

Subtract 2 times Row 1 from Row 3:

$$\begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 3 \\ 0 & 1 & 0 & -5 \end{bmatrix}$$

Subtract Row 2 from Row 3:

$$\begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & -8 \end{bmatrix}$$

Multiply Row 3 by $(-1/8)$:

$$\begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Matrix is now in REF with 3 leading ones, and so the rank of D is 3.

Activity: Bases and dimensions of matrix subspaces (U2-LO3, U2-LO4)

Let

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & -3 \\ 0 & 1 & -2 \\ 1 & 1 & -1 \end{bmatrix}$$

be the matrix we considered in the solution to Finding a basis for a span (U2-LO4).

What is $\dim(\text{row}(A))$?

The dimension of the row space of A is equal to the rank of A , which is 2 (the number of non-zero rows in the row echelon form for A , as computed in the solution to Finding a basis for a span (U2-LO4)).

Activity: Bases and dimensions of matrix subspaces (U2-LO3, U2-LO4)

Let

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & -3 \\ 0 & 1 & -2 \\ 1 & 1 & -1 \end{bmatrix}$$

be the matrix we considered in the solution to Finding a basis for a span (U2-LO4).

Find a basis for the column space of A .

The leading 1s in the row echelon form for A computed in Finding a basis for a span (U2-LO4) occur in the 1st and 2nd columns, so the 1st and 2nd columns of A form a basis for the column space, i.e.,

$$\left\{ \begin{bmatrix} 2 \\ 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 1 \\ 1 \end{bmatrix} \right\}$$

is a basis for $\text{col}(A)$.

Activity: Bases and dimensions of matrix subspaces (U2-LO3, U2-LO4)

Let

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & -3 \\ 0 & 1 & -2 \\ 1 & 1 & -1 \end{bmatrix}$$

be the matrix we considered in the solution to Finding a basis for a span (U2-LO4).

What is $\dim(\text{col}(A))$?

The dimension of the column space of A is two, since we constructed a basis with two elements in the previous task. Alternatively, the dimension of the column space is equal to the rank of A , which we also computed to be two.

Activity: Bases and dimensions of matrix subspaces (U2-LO3, U2-LO4)

Let

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & -3 \\ 0 & 1 & -2 \\ 1 & 1 & -1 \end{bmatrix}$$

be the matrix we considered in the solution to Finding a basis for a span (U2-LO4).

Find a basis for the nullspace of A .

The Elimination Scratchpad

To find a basis for the solutions to the equation $A\mathbf{x} = \mathbf{0}$, we use the row-echelon form computed in Finding a basis for a span (U2-LO4) and find basic solutions as in Basic solutions to homogeneous linear equations. The row-echelon

form gives the equations $x_1 + \frac{1}{2}x_2 = 0$ and $x_2 - 2x_3 = 0$. Setting the free variable $x_3 = 1$, we obtain $x_2 = 2$ and $x_1 = -1$, so the basic solution is

$$\begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}.$$

Thus $\left\{ \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} \right\}$ is a basis for the nullspace of A .

Definition

The nullity of a matrix A is equal to $\dim(\text{null}(A))$. It is denoted $\text{nullity}(A)$.

Rank counts independent directions transmitted by a matrix. Nullity counts independent input directions forgotten by a matrix. The next theorem relates these two dimensions.

Theorem: Rank-Nullity Theorem

Let A denote an $m \times n$ matrix of rank r . Then

- The $n - r$ basic solutions to the equation $Ax = 0$ provided by the Gaussian algorithm are a basis for $\text{null}(A)$, so $\dim(\text{null}(A)) = n - r$.

- $\dim(\text{im}(A)) = r$.
- $\dim(\text{im}(A)) + \dim(\text{null}(A)) = n$.
- $\text{rank}(A) + \text{nullity}(A) = n$.

Note

input dimensions = transmitted dimensions + forgotten dimensions

$$n = \text{rank}(A) + \text{nullity}(A)$$

We can now state one of the preview patterns from Section 2.1, introduced in Two inputs, one output (U2-LO1, U2-LO6), in the language of null spaces.

Activity: Same output from a null-space direction (U2-LO3, U2-LO6)

Suppose $A\mathbf{x} = \mathbf{y}$ and $A\mathbf{z} = \mathbf{0}$. Compute $A(\mathbf{x} + t\mathbf{z})$, where $t \in \mathbb{R}$. What does this say about uniqueness?

$$\begin{aligned} A(\mathbf{x} + t\mathbf{z}) &= A\mathbf{x} + tA\mathbf{z} \\ &= \mathbf{y} + t\mathbf{0} \\ &= \mathbf{y}. \end{aligned}$$

So every vector $\mathbf{x} + t\mathbf{z}$ gives the same output. A nonzero vector in the null space creates a whole line of inputs with the same output.

Activity: Revisiting the height map with rank and nullity (U2-LO3)

Return again to A map that forgets height (U2-LO1, U2-LO3). Let

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

Find the column space, null space, rank, and nullity of P . How does rank-nullity describe the height direction that P forgets?

The column space is the span of the columns of P . Its first two columns are the standard basis vectors of $\mathbb{R}^2$, and its third column is zero, which adds no new vectors to the span. Therefore

$$\text{col}(P) = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\} = \mathbb{R}^2,$$

so $\text{rank}(P) = 2$.

To find the null space, solve $P\mathbf{x} = \mathbf{0}$:

$$P \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Thus $x_1 = x_2 = 0$, while x_3 is free. Setting $x_3 = t$ gives every solution as

$$\mathbf{x} = \begin{bmatrix} 0 \\ 0 \\ t \end{bmatrix} = t \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad t \in \mathbb{R}.$$

Hence

$$\text{null}(P) = \text{span} \left\{ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\},$$

so $\text{nullity}(P) = 1$. Since P has three input coordinates,

$$\text{rank}(P) + \text{nullity}(P) = 2 + 1 = 3.$$

Rank-nullity says that the two output directions and the one forgotten height direction account for all three input directions.

Activity: Image and null-space bases from RREF (U2-LO3, U2-LO4)

Consider the matrix

$$B = \begin{bmatrix} 3 & -1 & 2 & 0 \\ -6 & 2 & -2 & 6 \end{bmatrix}.$$

You may use the fact that the reduced row echelon form of B is the

Activity: Image and null-space bases from RREF (U2-LO3, U2-LO4) (continued)

matrix

$$\begin{bmatrix} 1 & -1/3 & 0 & -2 \\ 0 & 0 & 1 & 3 \end{bmatrix}.$$

Find a basis for $\text{im}(B)$.

We use the fact that $\text{im}(B) = \text{col}(B)$. Using Rank Theorem, since the RREF in the referenced item has pivots in the 1st and 3rd columns, the column space of B , and thus the image of B , has a basis given by the 1st and 3rd columns of B , i.e., we have a basis

$$\left\{ \begin{bmatrix} 3 \\ -6 \end{bmatrix}, \begin{bmatrix} 2 \\ -2 \end{bmatrix} \right\}.$$

Activity: Image and null-space bases from RREF (U2-LO3, U2-LO4)

Consider the matrix

$$B = \begin{bmatrix} 3 & -1 & 2 & 0 \\ -6 & 2 & -2 & 6 \end{bmatrix}.$$

You may use the fact that the reduced row echelon form of B is the matrix

Activity: Image and null-space bases from RREF (U2-LO3, U2-LO4) (continued)

$$\begin{bmatrix} 1 & -1/3 & 0 & -2 \\ 0 & 0 & 1 & 3 \end{bmatrix}.$$

Find a basis for $\text{null}(B)$.

The Elimination Scratchpad

The 2nd and 4th columns of the matrix in the referenced item do not contain pivots, so x_2 and x_4 correspond to free variables in the equation, and solutions to $Ax = 0$ are specified by the two equations $x_1 = x_2/3 + 2x_4$ and $x_3 = -3x_4$. The two basis solutions give a basis for the nullspace, and thus we have a basis of the form

$$\left\{ \begin{bmatrix} 1/3 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ -3 \\ 1 \end{bmatrix} \right\}.$$

Redundant features in a data matrix

A data matrix can have redundant columns. This means one feature column is a linear combination of other feature columns.

Example: Redundant features in house data

Rows below represent houses. The three features are

first-level area, second-level area, total area,
measured in hundreds of square feet:

$$X = \begin{bmatrix} 9 & 7 & 16 \\ 11 & 9 & 20 \\ 14 & 0 & 14 \\ 8 & 8 & 16 \end{bmatrix}.$$

Let $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ be the columns of X . The third column satisfies

$$\mathbf{v}_3 = \mathbf{v}_1 + \mathbf{v}_2,$$

so the third feature carries no new independent direction. The dependence relation is

$$\mathbf{v}_1 + \mathbf{v}_2 - \mathbf{v}_3 = \mathbf{0},$$

or equivalently

$$X \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = \mathbf{0}.$$

The null-space direction

$$\mathbf{z} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

says that coefficient vectors differing by a multiple of $\mathbf{z}$ give the same prediction vector.

Activity: Redundant square-footage features (U2-LO6)

Use the data matrix

$$X = \begin{bmatrix} 9 & 7 & 16 \\ 11 & 9 & 20 \\ 14 & 0 & 14 \\ 8 & 8 & 16 \end{bmatrix}.$$

1. Compare the coefficient vectors

$$\mathbf{c} = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{c}' = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}.$$

Compute $X\mathbf{c}$ and $X\mathbf{c}'$.

2. The two coefficient vectors tell different stories. Why do they make the same predictions?
3. What warning does this give about asking which feature is most important?

We compute

$$X\mathbf{c} = \begin{bmatrix} 34 \\ 42 \\ 42 \\ 32 \end{bmatrix}, \quad X\mathbf{c}' = \begin{bmatrix} 34 \\ 42 \\ 42 \\ 32 \end{bmatrix}.$$

The coefficient vectors are different, but they

differ by the null-space direction:

$$\mathbf{c} - \mathbf{c}' = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}.$$

The third column of X is the sum of the first two, so

$$X(\mathbf{c} - \mathbf{c}') = \mathbf{0}.$$

By linearity, $X\mathbf{c} - X\mathbf{c}' = \mathbf{0}$, hence $X\mathbf{c} = X\mathbf{c}'$.

The predictions are the same, but the coefficient interpretations are different. One coefficient vector puts weight on first-level and second-level area separately. The other puts weight on first-level area and total area. With redundant features, individual coefficient importance is not uniquely determined by the data matrix.

Further feature redundancy examples include:

- first-level area, second-level area, and total area;
- minutes and seconds, where seconds = 60 · minutes;
- height in inches and height in centimeters, where centimeters =

2.54 · inches;

- subtotal, tax, tip, and total bill, where $\text{total} = \text{subtotal} + \text{tax} + \text{tip}$;
- exam parts and exam total, where $\text{total} = \text{part 1} + \text{part 2} + \text{part 3}$.

These are linear redundancies. Not every relationship between features is linear. For example, rectangle area is width times height, not a linear combination of width and height.

Activity: Auditing redundant features in code (U2-LO3, U2-LO6)

The matrix below stores the house features from Redundant square-footage features (U2-LO6).

```
import numpy as np

X = np.array([
    [9, 7, 16],
    [11, 9, 20],
    [14, 0, 14],
    [8, 8, 16],
])

z = np.array([1, 1, -1])
c = np.array([3, 1, 0])
c_alt = np.array([2, 0, 1])

rank = int(np.linalg.matrix_rank(X))

X @ z, rank, X @ c, X @ c_alt
```

Here `c_alt` stores the vector c' .

Output:

```
(array([0, 0, 0, 0]),
2,
```

Activity: Auditing redundant features in code (U2-LO3, U2-LO6) (continued)

```
array([34, 42, 42, 32]),  
array([34, 42, 42, 32]))
```

1. What does the zero output mean?
2. What does the rank say about the three feature columns?
3. Why do Xc and Xc' agree?
4. What does this say about interpreting coefficients when features are redundant?

The zero output means that z is a null-space direction. The rank is 2, so the three feature columns contain only two independent directions.

The vectors c and c' differ by z , and $Xz = 0$. Therefore they give the same prediction vector. When features are redundant, different coefficient vectors can make the same predictions. Individual coefficients may not have a unique interpretation.

Activity: A difference matrix forgets level (U2-LO3)

Recall the difference matrix from A difference matrix (U1-LO4, U1-LO5). For four input coordinates, let

$$D = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}.$$

Compute $D \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$. What input direction is forgotten?

We get

$$D \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = \mathbf{0}.$$

The matrix records consecutive differences. Adding the same constant to every entry does not change the differences, so the constant direction $(1, 1, 1, 1)$ is forgotten.

Activity: Reading the difference matrix in code (U2-LO3, U2-LO6)

```
D = np.array([
    [-1, 1, 0, 0],
    [0, -1, 1, 0],
    [0, 0, -1, 1],
])

x = np.array([2, 5, 9, 10])
ones = np.ones(4, dtype=int)

D @ x, D @ (x + 10*ones), D @ ones
```

Output:

```
(array([3, 4, 1]),
array([3, 4, 1]),
array([0, 0, 0]))
```

1. What does $D @ x$ measure?
2. Why do the first two outputs agree?
3. What does $D @ \text{ones}$ say about the map?

The product $D @ x$ measures consecutive differences:

$$5 - 2 = 3, \quad 9 - 5 = 4, \quad 10 - 9 = 1.$$

Adding 10 to every entry changes the level but not the differences, so the first two outputs agree.

The output $D @ \text{ones}$ is 0 , so the constant direction is a null-space direction. The difference matrix forgets level.

Redundant features create null-space directions. If $Xz = 0$, then

$$X(c + tz) = Xc$$

for every scalar t . Thus many coefficient vectors can produce the same prediction vector. Rank counts how many independent feature directions remain; nullity counts how many independent coefficient changes are invisible to the data matrix.